Assignment 3: Question 4, part 1

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression

# Features:
# X[:,0] = Income
# X[:,1] = Loan

X = np.array([
    [40,10],
    [45,15],
    [50,12],
    [55,20],
    [60,18],
    [65,22],
    [70,24],
    [75,20],
    [80,25],
    [85,30],
    [90,35],
    [95,40]
])

# 0 = No Default
# 1 = Default
y = np.array([0,0,0,0,0,1,1,1,1,1,1,1])


# Train Logistic Regression
model = LogisticRegression()

model.fit(X,y)

# Prediction
customer = np.array([[80,25]])

prediction = model.predict(customer)
probability = model.predict_proba(customer)

print("Prediction:", prediction[0])
print("Probability:", probability)

print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
```

```
Prediction: 1
Probability: [[4.94867786e-06 9.99995051e-01]]
Intercept: [-44.68298282]
Coefficients: [[0.58906753 0.39095864]]
```

Assignment 3: Question 4, part 2

In [3]:
```
import numpy as np
import matplotlib.pyplot as plt
```

```python
# Generate z values
z = np.linspace(-10, 10, 200)

# Sigmoid function
sigmoid = 1 / (1 + np.exp(-z))

# Plot
plt.figure(figsize=(8,5))
plt.plot(z, sigmoid, linewidth=2)

# Decision threshold
plt.axhline(0.5, linestyle='--', color='red', label='Threshold = 0.5')

plt.xlabel("Linear Score (z)")
plt.ylabel("Probability")
plt.title("Sigmoid Function")
plt.grid(True)
plt.legend()

plt.tight_layout()
plt.show()
```